Nguyễn Mạnh Hùng

Backend Engineer

Backend Engineer with 3 years building distributed systems serving 5,000+ devices across 5 countries. Specializing in Java/Spring Boot, Kafka, and Kubernetes, with hands-on AI integration experience shipping computer vision pipelines to enterprise clients.

SKILLS

Core Expertise	Java 17 · Spring Boot · Kafka · Event-Driven Architecture · Microservices · Distributed Systems
Infrastructure	Kubernetes · Helm · Docker · Jenkins · Prometheus · Grafana · ELK/OpenSearch · Keycloak · NGINX Ingress · SonarQube
Data & Messaging	PostgreSQL · MariaDB · Redis · Elasticsearch · Kafka (migration + production ops)
AI Integration	FastAPI · RAG pipelines · OpenAI API · On-premise GPU inference · Computer Vision (Pytorch, ONNX)
Languages	Java · Golang · Python · TypeScript
Tools	Git · Jira · Confluence · Robot Framework · WireGuard · n8n

CONTACT

- nmhung.works@gmail.com
- Ho Chi Minh City, Vietnam
- +84 947 339 718
- github.com/jinergenkai
- linkedin.com/in/jinergenkai
- Jiner Blog

EDUCATION

• Bachelor of IT *2020 - 2024*

FPT University, HCMC

GPA 3.2/4.0 • 100% Scholarship

• High School for Gifted *2017 - 2020*

Nguyen Du HS (Informatics), Dak Lak

AWARDS

- **ICPC 2023** - Third Prize
  Vietnam Southern Region
- **Meta Hacker Cup 2025**
  Round 2 - Top 12% Global
- **ICPC 2021** - Honorable Mention
  Vietnam National
- **National Excellent Student 2020**
  Informatics - Honorable Mention

EXPERIENCE

TMA Solutions – Backend Engineer *Apr 2024 - Present*

→ Nokia Altiplano – Access Controller Team

Designed campaign management system within Nokia Altiplano platform, orchestrating firmware upgrades for 5,000+ telecom devices across distributed optical network infrastructure spanning 5 countries.

• Designed distributed firmware upgrade orchestration using NETCONF/YANG with dual-bank architecture, parallel execution across 3 regions, circuit breaker patterns, and automated rollback
• Diagnosed critical Kafka consumer lag incident caused by fixed partition key – implemented distributed key strategy restoring real-time processing for 20+ microservices
• Owned Build Police role for 20+ member cross-regional team (Vietnam/India/China/Greece/Finland) – Jenkins CI/CD health, flaky test reduction via Robot Framework, code quality enforcement via SonarQube
• Migrated event streaming from JMS to Kafka, enabling scalable inter-domain communication across AC and IBN services
• Resolved 15+ CVE security tickets using Syft SBOM analysis across containerized microservices

Tech: Java 17, Spring Boot, Kafka, Helm, Kubernetes, NGINX Ingress, Jenkins, MariaDB, Redis ELK/OpenSearch, Prometheus, Grafana, Keycloak, NETCONF/YANG

→ Agentic Vision – AI-Powered Security Platform (dedicated assignment)

Architected backend for AI-powered security monitoring platform with real-time video streaming and escalating alert workflows, processing 100+ camera feeds. Deployed on-premise for enterprise clients including Novaland.

• Built FastAPI streaming backend with WebSocket for real-time 100+ IP camera monitoring, automated alert pipelines reducing manual workload by 67%
• Designed escalating alert workflow engine: user-configurable trigger chains (1min → 5min intervals, progressive severity) routing to mail/SMS recipients – enabling custom security response automation
• Implemented token-aware prompt optimization for OpenAI API, balancing inference accuracy vs cost with response caching to cut redundant calls

Tech: Python, FastAPI (StreamResponse), WebSocket, Redis, OpenAI API, Docker

FPT Software – Software Engineer *Aug 2023 - Apr 2024*

Developed medical imaging software for ultrasound analysis – performance optimization and API design for international healthcare clients.

• Optimized imaging application by offloading heavy I/O to background threads, reducing startup time by 20%
• Restructured backend API response formats for efficient high-throughput medical data delivery

Tech Stack: C++, C#, ASP.NET Core

PROJECTS

ThuleHT – E-commerce Platform

2025 - Present

→ thuleht.com

Go microservices backend for cosmetics e-commerce platform with integrated search, messaging, and chatbot infrastructure.

• Built notification service with multi-channel delivery (Zalo OA API, email) – order updates, promotional messaging, and Zalo Official Account registration/verification
• Developed search service using Elasticsearch with Vietnamese language support for product discovery
• Implemented chatbot with MCP-based tool integration for automated customer inquiries and order tracking

Golang, Elasticsearch, Zalo OA API, Docker, PostgreSQL

J Post Image — AI Content Pipeline

2025 - Present

→ t.me/jiner_screen_bot · Try it: 5 free credits per telegram user

End-to-end AI content pipeline: local GPU image/video generation → automated posting → audience engagement, orchestrated via Telegram bot.

• Architected on-premise GPU inference with WireGuard VPN tunneling through VPS hub – serving ZImage/Qwen models from local RTX via Telegram bot interface
• Built automated distribution: ComfyUI generation → Postiz scheduling → multi-platform posting, with Supabase storage and FastAPI orchestration
• Designed model-agnostic workflow enabling rapid swap between FLUX, SDXL, and video generation models without pipeline reconfiguration

Python, FastAPI, ComfyUI, WireGuard, Supabase, Telegram Bot API, Docker, n8n

Huynh Hanh – Management System

2024 - Present

→ huynhhanh.com · github.com/huynhhanh-ms

Multi-service platform for agricultural operations with Java and Go microservices backends.

• Built Java storage service with RAG pipeline for intelligent document search across operational records
• Developed Go land management service with geospatial data handling for multi-site property tracking
• Integrated hardware data sync via Web Serial API for weighing stations with automated receipt generation

Java, Spring Boot, Golang, PostgreSQL, Docker, RAG